

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, AUTUMN SEMESTER 2010-2011

MACHINE LEARNING

Time allowed: Two Hours

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer All Questions

Only silent, self contained calculators with a Single-Line Display or Dual-Line Display are permitted in this examination

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

Step0: Treat the learning system as a black box

Step1: Collect the training examples(Experience E)

1. so-called learning from examples

Step2: Representing Experience:

1. Choose a representation scheme for the experience / examples

1. use the n-d vector to represent experience called feature vector. $X=(x_1, x_2, x_3, \dots, x_n)$

2. Need a corresponding vector D, which will record the knowledge about X

3. The experience E is a pair of vectors $E=(X,D)$

4. Assume our system is to recognise 10 digits only, then D can be a 10-d binary vector; each correspond to one of the digits

$D=(d_0, d_1, \dots, d_9)$ e.g. if X is digit 5, then $d_5=1$; all others = 0; $D=(0,0,0,0,0,1,0,0,0,0)$

Step 3: Choose a representation for black box

1. choose a function F to approximate the black box. for given X, the value of F will give the classification of X. There are considerable flexibilities in choosing F

2. F will be a function of some adjustable parameters, or weights, $W = (w_1, w_2, w_3, \dots, w_N)$, which the learning algorithm can modify or learn

Step 4: Learning/adjusting the weights

1. We need a learning algorithm to adjust the weights such that the experience/prior knowledge from the training data can be learned into the system: $E=(X,D)$. $F(W,X) = D$ Error= $\|D-F(W,X)\|$

Step 5: Use / Test the system

1. Once learning is completed, all parameters are fixed. An unknown input X is presented to the system, the system computes its answer according to $F(W,X)$

Generalization refers to the ability of a machine learning model to perform well on unseen test data, rather than only on the training data. A model with good generalization learns the underlying patterns of the data so that the training and test errors are both low and close to each other.

Over-fitting occurs when a model fits the training data too closely, including noise and sampling errors. As a result, the model achieves very low training error but performs poorly on unseen test data, leading to a large gap between training and test performance. Over-fitting is often caused by overly complex models with too many parameters.

One possible solution to avoid over-fitting is regularization, such as weight decay (L2 regularization). Weight decay penalizes large weights during training, encouraging smoother models and reducing sensitivity to noise, thereby improving generalization performance.

Question 1

- (a) Using handwritten digit recognition as an example, list the major steps of building a machine learning system and for each step discuss what it is involved in detail. Use a specific or a generic learning model as appropriate to illustrate your answer. (10 marks)
- (b) Discuss the concepts of **generalization** and **over-fitting** in the context of machine learning and give one possible solution to avoid over-fitting. (10 marks)

Question 2

- (a) Design a single layer perceptron that will classify the following 3-d inputs into two classes.

Input	Output
(1, 0, 0,)	Class 1
(0, 1, 0)	Class 1
(1, 1, 0)	Class 1
(0, 0, 2)	Class 2
(0, 0, 1)	Class 2

Your answer should include:

- (i) A diagram of the perceptron.
 - (ii) The values of the connection weights and the threshold or bias connection.
 - (iii) The equation of the perceptron’s decision boundary.
 - (iv) A verification of the correctness of your design. (10 marks)
- (b) A network consists of two ADLINE units, N1 and N2 is shown in Fig. Q2(b).
- (i) Derive a delta training rule for all the weights, you must show the derivation process. (5 marks)
 - (ii) Assuming the weights have following values $w_0= w_1= w_2= w_3= w_4= 0.1$, using a training rate of 0.1, show the new values of w_0 and w_4 after presenting a training sample $x_1= 0.5, x_2=0$ with a desired output $d = 0.5$ (5 marks)

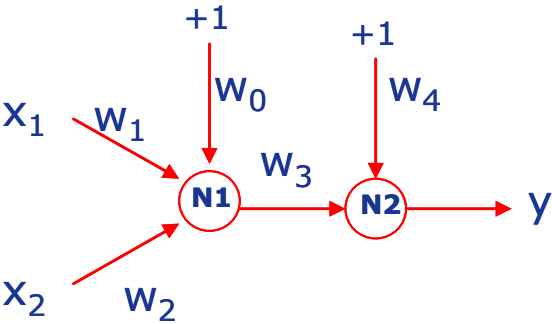


Fig. Q2(b)

4a Both the perceptron and linear SVM classifiers use a hyperplane to separate linearly separable classes. However, their training algorithms are fundamentally different.

The perceptron algorithm aims to find any separating hyperplane and updates its weights only when a training example is misclassified. It does not explicitly maximize the margin, and its solution depends on the order of the training data. In contrast, a linear SVM explicitly maximizes the margin between the two classes by solving a convex optimization problem using hinge loss. Even correctly classified samples that lie within the margin can influence the training process. As a result, linear SVM finds a unique optimal hyperplane and generally exhibits better generalization performance than the perceptron.

Question 3

In an object recognition task, it is known that the objects come from one of two classes, C_1 or C_2 . Each instance of an object X has four features, $X = (x_1, x_2, x_3, x_4)$. An experiment has collected 14 instances and their feature values and classification are shown in the following table

$X = (x_1, x_2, x_3, x_4)$.				Classification
x_1	x_2	x_3	x_4	C
1	1	1	1	C_1
1	1	1	2	C_1
2	1	1	1	C_2
3	2	1	1	C_2
3	3	2	1	C_2
3	3	2	2	C_1
2	3	2	2	C_2
1	2	2	1	C_1
1	3	2	1	C_2
3	2	2	1	C_2
1	2	2	2	C_2
2	2	1	2	C_2
2	1	2	1	C_2
3	2	1	2	C_1
1	3	1	2	=?

Using the data with known classification, we can build a Naïve Bayesian classifier to classify the unknown instance $X = (1, 2, 1, 2)$ in the last row of the table. According to the Naïve Bayesian classifier, the classification of the unknown instance, C , can be expressed as

$$C = \arg \max_{C \in \{C_1, C_2\}} P(C)P(x_1 = 1 | C)P(x_2 = 2 | C)P(x_3 = 1 | C)P(x_4 = 2 | C)$$

Which class will the instance $X = (1, 2, 1, 2)$ be classified into? You must show the detail of the working process. An answer without the working process will not receive any marks.

4b False

(20 marks)

Although only support vectors determine the final decision hyperplane of an SVM, all training samples are required during the training process to identify these support vectors. Therefore, non-support vectors cannot be ignored in SVM training.

Question 4

(a) The perceptron and linear SVM classifier both use a pyherplane to separate two linearly separable classes. Compare and contrast their respective training algorithms.

(8 marks)

(b) The Support Vectors are the only training samples that are useful for determining the decision hyperplane of the SVM, hence the rest training samples can be ignored in SVM training. True or false? Discuss.

4c Support Vector Machines (SVMs) handle linearly non-separable data mainly through soft margin classification and kernel methods.

1) Soft Margin SVM

(6 marks)

When data are not perfectly linearly separable, SVM introduces slack variables to allow some training samples to violate the margin constraints, even be misclassified. This leads to the soft margin SVM which balances two objectives:

- maximizing the margin between classes, and
- minimizing classification errors.

(6 marks)

A regularization parameter C controls this trade-off. A large C penalizes misclassification more heavily and leads to a smaller margin, while a smaller C allows more violations but improves generalization.

2) Kernel Trick

For data that are inherently non-linearly separable in the original input space, SVM applies the kernel trick. A kernel function implicitly maps the input data into a higher-dimensional feature space where the data may become linearly separable. Common kernels include the polynomial kernel and the radial basis function (RBF) kernel.

By computing inner products in the transformed space without explicitly performing the mapping, SVM can efficiently construct a non-linear decision boundary in the original input space.

One-sentence Summary (Exam-friendly)

SVMs handle linearly non-separable data by allowing margin violations through soft margins and by using kernel functions to construct non-linear decision boundaries in higher-dimensional feature spaces.

Turn over

Answer: Dataset X achieves more effective dimensionality reduction.

Why (PCA intuition from lectures):

- PCA is most effective when features are highly correlated / redundant, because most variance lies along one dominant direction (the first principal component), and the remaining component(s) capture relatively little extra information.
- In X, the large covariance (0.85) indicates that x_1 and x_2 move together strongly $\rightarrow$ the data are close to a 1D structure in 2D space $\rightarrow$ one principal component can summarize the data well.
- In Y, covariance is small (0.1) $\rightarrow y_1$ and y_2 are almost independent $\rightarrow$ variance is spread across both axes $\rightarrow$ you generally need both principal components to retain most information, so dimensionality reduction is less effective.

Exam-friendly one-liner:

PCA reduces dimension best when variables are strongly correlated (since x_1 and x_2 show much higher covariance (redundancy) than y_1 and y_2 , X can be reduced more effectively (often to 1D) than Y.
(a) Experiment 1 has produced N 2d data points $\{X(i) = (x_1(i), x_2(i))\}$, $i = 1, 2, \dots, N$. Experiment 2 has produced M 2d data points $\{Y(i) = (y_1(i), y_2(i))\}$, $i = 1, 2, \dots, M$. The covariance matrices of these two sets of data, S_x and S_y , are

$$S_x = \begin{bmatrix} 0.7 & 0.85 \\ 0.85 & 0.5 \end{bmatrix} \qquad S_y = \begin{bmatrix} 0.8 & 0.1 \\ 0.1 & 0.6 \end{bmatrix}$$

- (i) What can be inferred about the relations between x_1 and x_2 ; and that between y_1 and y_2 .
(5 marks)
- (ii) If Principal Component Analysis (PCA) is applied to reduce the dimensionality of X and Y with the aim of reducing the dimensionality of the measurements, which dataset can achieve more effective dimensionality reduction and why.
(5 marks)
- (b) An experiment has produced the following 3d feature vectors $X = (x_1, x_2, x_3)$ belonging to two classes. Design a decision tree classifier to class an unknown feature vector $X = (1, 2, 1)$.

$X = (x_1, x_2, x_3)$			
x_1	x_2	x_3	Classes
1	1	1	1
1	1	1	2
1	1	1	1
2	1	1	2
2	1	2	1
2	2	2	2
2	2	2	1
2	2	1	2
1	2	2	2
1	1	2	1
1	2	1	= ?

(10 marks)